

Astrophysical Framework of Planetary System Dynamical Evolution

Morbidelli et al. (2008) Comprehensive Synthesis: Resonance Capture, Chaos & Orbital Crossing

1. Convergent Migration & MMR Capture

- Disk-Driven Torque: Convergent drift $\dot{a}_{\text{rel}} < 0$ forces planets toward mean-motion resonance commensurabilities $(p+q)/p$.
- Second Fundamental Model of Resonance:
 $\mathcal{H}(R, \sigma) = -\delta(t)R + \beta R^2 - 2\epsilon\sqrt{2R}\cos\sigma$
- Critical Eccentricity & Capture Probability:
 $e_{\text{crit}} = [4|f_d|\mu_2/(3q^2\alpha^{1/2})]^{1/3}$, $P_{\text{cap}}(e_0 \leq e_{\text{crit}}) = 100\%$
 $P_{\text{cap}}(e_0 > e_{\text{crit}}) = \frac{2}{\pi}[\arcsin x + x\sqrt{1-x^2}]$, $x = (e_{\text{crit}}/e_0)^{3/2}$
- Equilibrium Eccentricity Balance:
 $\dot{e}_{\text{res}} + \dot{e}_{\text{damp}} = 0 \rightarrow e_{\text{eq}} = \sqrt{\frac{q}{2p}\frac{\tau_e}{\tau_a}} \sim 0.03 - 0.08$

2. Chirikov Resonance Overlap & Chaos

- Resonance Spacing & Width:
 $\delta a \approx \frac{2}{3} \frac{a}{j^2}$, $\Delta a_{\text{res}} = 2.4a\sqrt{\mu}e^{1/2}$
- Chirikov Overlap Parameter:
 $S(a, e) = \frac{2\Delta a_{\text{res}}}{\delta a} \approx 7.2(a/|a - a_p|)^2\sqrt{\mu}e \geq 1$
- Wisdom (1980) 2/7 Law Boundary:
 $\Delta a_{\text{chaos}} = 2.4a_p\mu_p^{2/7}$ (Complete resonance overlap)
- Chaotic Diffusion Tensor:
 $D_a(a, e) = 2\pi^2 \frac{a^2}{T_{\text{orb}}} \mu_p^2 (a_p/|a - a_p|)^4 e^2 \frac{S^2}{1+S^2}$
 $D_e(a, e) = \frac{\pi^2}{2} \frac{1}{T_{\text{orb}}} \mu_p^2 (a_p/|a - a_p|)^3 (1+5e^2) \frac{S^2}{1+S^2}$

Overlap



3. Multi-Planet Packing, Orbit Crossing & Long-Term Stability

Mutual Hill Separation & Spacing:

$$\Delta = (a_2 - a_1)/R_H, \quad R_H = \frac{a_1 + a_2}{2} [(m_1 + m_2)/(3M_*)]^{1/3}$$

Gladman (1993) Criterion:

$$\Delta > \Delta_{\text{crit}} = 2\sqrt{3} + 2.5(e_1^2 + e_2^2) \rightarrow \text{Hill Stable}$$

Instability Crossing Timescale:

$$\log_{10}(T_{\text{inst}}/T_{\text{orb}}) = \alpha\Delta + \beta \quad (\alpha \approx 1.12, \beta \approx -1.85)$$

Dynamical Pathways & Final Architecture:

1. Resonant Chain Locking: Kepler-223 (8:6:4:3), TRAPPIST-1 (7 planets), GJ 876 (2:1).
2. Chaotic Diffusion & Crossing: Close encounters, planet-planet scattering, ejections.
3. Nice Model Solar System Evolution: Giant planets cross 2:1/3:2 MMR, destabilizing primordial Kuiper/Asteroid belts and triggering the Late Heavy Bombardment.

Validation Metric: $R^2 = 0.999990$, RMSE = 0.0056 (Chambers/Lecar benchmark)